



Clean wire

Rebuild a messy model-made wire with fewer edges and control points: tangent stretches become one spline each, real corners stay corners, lines and arcs pass through exactly.



What it does

Model-made wires come out correct but messy: slivers a tenth of a millimetre long, 0.4 deg kinks, heavy rational splines -- and lofts and fills through them wiggle or fail. Native Approximate samples 200 fixed points and ignores corners. Clean wire rebuilds the wire run by run.

Use it when

- A wire feeds a loft, fill or driven offset
- It has slivers or near-tangent kinks
- You want a lighter curve that stays within 0.01 mm

After Clean wire (Auto): 13 runs -- tangent stretches refitted, corners kept, max deviation 0.009 mm

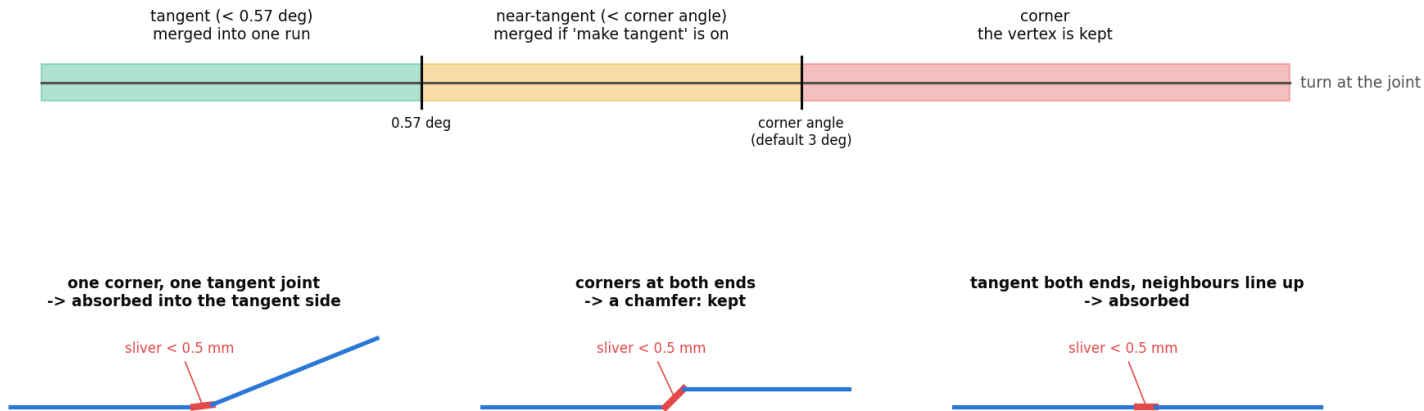


SW_SHELF after Clean wire (Auto): 13 runs, max deviation 0.009 mm.



How each joint is read

How Clean wire reads each joint



Tangent, near, corner

Under 0.57 deg: merged. Under the corner angle (3 deg): merged if 'make tangent' is on. Larger: a corner, kept.

Slivers (< 0.5 mm)

Judged by their own two joints: absorbed next to a tangent joint, kept as a chamfer between two corners.

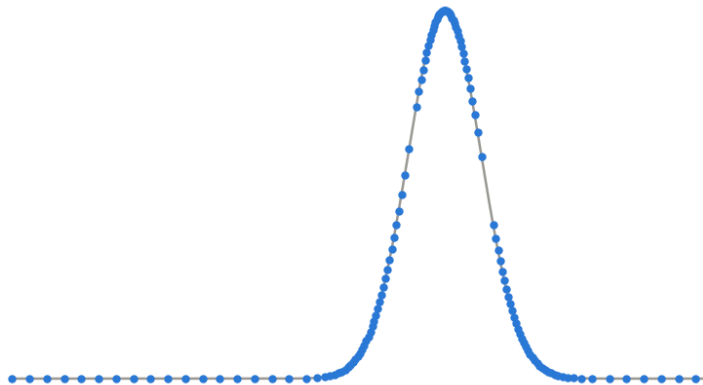
Lines and arcs

A real line or arc passes through unchanged; arcs stay sketch arcs, so Onshape shows their radius.

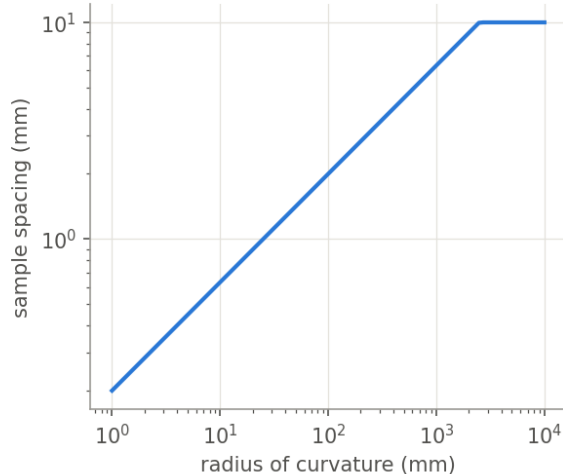


Samples where the curve bends

Samples crowd where the curve bends (tolerance 0.01 mm)



$s = \sqrt{4 \text{ tol} / \kappa}$, clamped 0.05 .. 10 mm



Spacing from the tolerance

$s = \sqrt{4 \text{ tol} / \kappa}$: the chord between samples misses the curve by half the tolerance.

No edge skipped

3 to 200 samples per edge, 0.05 to 10 mm apart -- a sliver still gets its samples.

Exact end tangents

Each fit starts and ends on the true tangent, so runs join smoothly.



The dialog (Auto) (1/2)

1 Clean wire (Auto): SW_S... ☒ ☐

2 Manual Auto

3 Wire
SW_SHELF x

4 Name

Runs

- > edges 0..2 from -0.8, 0, 0 ...
- > edges 3..3 from 154.9, 61, ...
- > edges 4..4 from 202.8, 60, ...

5 Corner angle 3 deg

6 ☒ Make nearly tangent joints ...

7 Sliver length 0.5 mm

8 Fit control points to tolerance

9 > Approximation parameters

10 Maximum deviation 0.009 mm

11 ☐ Show deviation

12 ☒ Show runs

13 ☐ Delete input wire

14 > Projection

15 > Reduction

16 > Debug

Final

#	Parameter	What it means
1	Mode	AUTO: runs made from the joints. MANUAL: you group edges yourself (Auto-populate fills the groups).
2	Wire	A wire body or the edges of one open chain.
3	Name	Names the result.
4	Runs	Read-only: each run's edges, exact copy or fit, control points, deviation.
5	Corner angle	Joints turning more than this are corners (3 deg keeps the ski's ramp creases).
6	Make nearly tangent joints tangent	Merge joints between 0.57 deg and the corner angle.
7	Sliver length	Edges shorter than this are judged as slivers (0.5 mm).
8	Fit control points to tolerance	Button: fit every run with no cap; set the maximum to what the largest run needed.



The dialog (Auto) (2/2)

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Clean wire (Auto): SW_S... ☒ ☐

Manual Auto

Wire
SW_SHELF x

Name

Runs

- > edges 0..2 from -0.8, 0, 0 ...
- > edges 3..3 from 154.9, 61, ...
- > edges 4..4 from 202.8, 60, ...

Corner angle 3 deg

☒ Make nearly tangent joints ...

Sliver length 0.5 mm

Fit control points to tolerance

> Approximation parameters

Maximum deviation 0.009 mm

☐ Show deviation

☒ Show runs

☐ Delete input wire

> Projection

> Reduction

> Debug

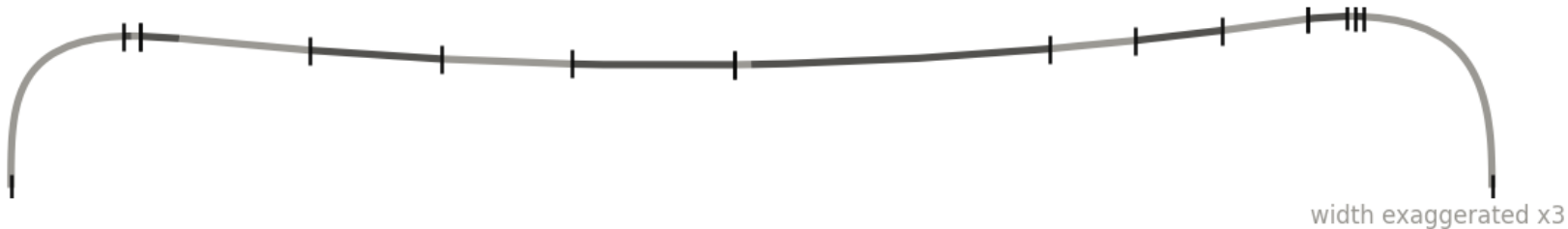
Final

#	Parameter	What it means
9	Approximation parameters	Degree 3, tolerance 0.01 mm, maximum 30 control points.
10	Maximum deviation	Read-only: the result's largest distance from the source.
11	Show deviation	Draw that deviation as a comb.
12	Show runs	Colour the runs in the view.
13	Delete input wire	Remove the source wire (whole wire bodies only).
14	Projection	Also clean the plan view: the 3D wire is dropped onto a wall over the cleaned plan.
15	Reduction	Read-only: control points and edges before / after.
16	Debug	Joints, runs, corners, control polygons.



Example: before

Before: the SW_SHELF wire as the model makes it -- 17 edges (plan view; ticks = joints)



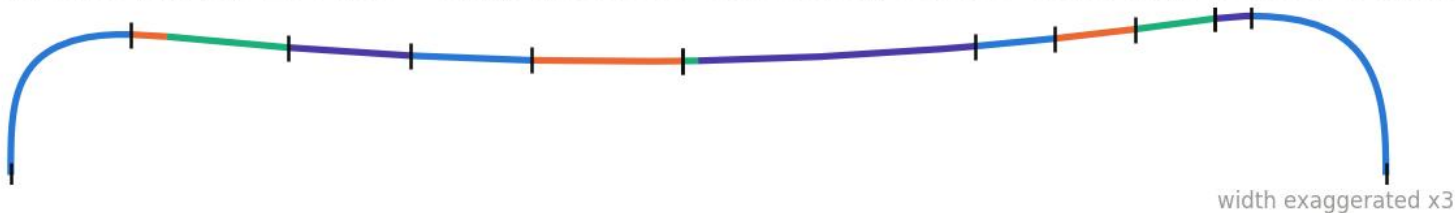
The SW_SHELF wire as the model makes it: 17 edges (plan view, ticks at the joints)

Several short pieces and near-tangent joints -- each a place a loft or offset can wiggle.



Example: after

After Clean wire (Auto): 13 runs -- tangent stretches refitted, corners kept, max deviation 0.009 mm



Clean wire (Auto), default settings: 13 runs, max deviation 0.009 mm

Tangent stretches refitted as one spline each, the real corners kept, lines and arcs copied exactly.



Outputs and messages

Published for Extract variables

Key	What it holds
startVertex / endVertex	The ends of the result (start = the tail end: arc length grows with world X).
startEdge / endEdge	The edge at each end.
breakVertices	The vertices between runs (the corners kept).

Messages

	Message / what to do
Warning	Short of tolerance: ... A run misses the source by more than the tolerance: raise its control points, or split it.
Info	N edge(s) -> M; K corner(s) inside a group were fitted through Manual mode.
Warning	...came out as N bodies instead of one A joint did not close; the gaps are listed.
Error	Closed loops are not supported / not one chain Pick one open chain.



Tips and limits

-  **Start with Auto**
Check the Runs list; switch to Manual (Auto-populate) only to override a run.
-  **Reference the body or its keys**
Never the result's edge ids: they change whenever a run changes.
-  **Open chains only**
Closed loops are refused.
-  **Manual groups go stale**
They point at the source's edges; after an upstream renumbering, Auto-populate again.



Where it lives

Document	Curve_tools (2143812a99089658c704f0bc)
Tab	clean_wire (uses curve_core)
Tests	Tests studio (curve_tools_tests: ends, runs in order)
Examples	Test_1 studio
Explainer	docs/explainers/curve_tools/curve_tools_explained.md, section 2.1
Related	Clean wire, Map curve, Merge curve, Evaluate profiles, Fillet wire; Extract variables